

NABANITA KAR



AEROSPACE ENGINEER

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An enthusiastic aerospace engineering graduate, seeking job, in a challenging area that focuses on the working of engines (propulsion), material science, and related areas, resulting in organizational and knowledgeable growth.

EDUCATION

DEGREE	INSTITUTE NAME	PERCENTAGE/CGPA	PASSING YEAR
B. Tech Aerospace	Hindustan Institute of Technology and Science	8.84	2021
12 th (HSC)	Khalsa Model Senior Secondary School	70	2017
10 th (SSLC)	Khalsa Model Senior Secondary School	8.8	2015

WORK EXPERIENCE

[HINDUSTAN AERONAUTICS LIMITED, BANGALORE] - [GRADUATE APPRENTICE TRAINEE]

Aircraft Research & Design Centre (ARDC)- Stress Group- July 2023- Current

- Process planning i.e., preparation of stage operation drawing using UG/NX-11 and AutoCAD drafting.
- Knowledge of Aerospace materials and processes.
- Study of aerospace manufacturing and assembly planning.
- Manufacturing planning and authoring assembly work.
- Understanding the geometry of the component, geometry cleanup techniques and deciding the type of mesh to be used.
- Generating 1D, 2D and 3D finite element model using MSC PATRAN.
- Knowledge on different methods of applying constraints and applying boundary conditions.
- Linear static analysis using MSC NASTRAN.
- Hand calculation for failure prediction.
- Size, shape and material optimization techniques.

CERTIFICATIONS

- Internship at HAL-Aircraft division on 2018.
- Participated in NATIONAL STUDENTS SPACE CHALLENGE organized by IIT KHARAGPUR on 2018.
- Participated in the National Workshop on challenges and modern techniques in realizing complex infrastructure by ISRO-SHAR, Sriharikota on 2018.
- Participated in workshop on Machine Learning organized by IIT HYDERABAD on 2020.
- Completed basic C programming course by Coursera.

SOFTWARE PROFICIENCY

- AutoCAD
- GD&T
- NX-11
- CATIA V5
- FEA- MSC PATRAN/ MSC NASTRAN
- General purpose- MS Office (Word, Excel, PowerPoint)

PROJECTS

SENSITIVITY ANALYSIS OF AN AIRCRAFT BRACKET

- Performed sensitivity analysis of an aircraft's bracket.
- Appropriate mesh and global edge length was decided for further analysis.

STRESS ANALYSIS OF OUTBOARD PYLON

- Analysis was carried out on outboard pylon to carry different kind of missiles.
- Grid point forces (GPF) were extracted.
- Failure prediction using hand calculation performed.
- Design modification was proposed which is feasible for production and without addition of mass.

STRESS ANALYSIS OF LUG

- Performed stress analysis of lug model for different loading condition.
- Different modes and types of failure were studied.

FINAL YEAR PROJECT

Designed bumps on the airfoil using CATIA V5 and UG/NX and done the flow analysis for various Mach numbers. Also, analyzed the boundary layer diversion.

HOBBIES

- Art and Craft
- Dance
- Music
- Kitchen Gardening

PERSONAL DETAILS

- D.O.B: 17-03-1999
- Languages Known: English
Hindi
Bengali